

Table 1: Direct comparison of classical and deep mixture-classification results on the original and PT2 real-mixture datasets

Setup	Family	Evaluation protocol	Existing exact	Existing Micro-F1	Existing avg. pred. chemicals	PT2 exact	PT2 Micro-F1
Pair NNLS + family near-tie fallback	Classical	Global baseline fallback + family margin 0.002	0.984	0.992	2.00	1.000	1.000
Pair NNLS + selective low-baseline fallback	Classical	Global rerank on baseline-heavy pair (thr 0.025, alt 0.005, margin 0.05)	0.969	0.984	2.00	1.000	1.000
Replicate-aware pair NNLS, baseline_corrected	Classical	Constant baseline atom, extra reps 9	0.907	0.953	2.00	1.000	1.000
Cardinality-adaptive NNLS	Classical	Support size 1/2/3 with calibrated penalty	0.900	0.969	2.05	0.889	0.973
Pair NNLS + nuisance baseline atom, baseline_corrected	Classical	Bernstein baseline atoms, degree 0	0.888	0.944	2.00	1.000	1.000
Exhaustive pair NNLS, baseline_corrected	Classical	All binary pairs, class-mean dictionary	0.864	0.932	2.00	1.000	1.000
Non-negative elastic net, baseline_corrected	Classical	Full-library positive ElasticNet	0.669	0.868	2.13	0.889	0.971
Non-negative elastic net, raw	Classical	Full-library positive ElasticNet	0.567	0.766	1.93	0.981	0.995
Exhaustive pair NNLS, raw	Classical	All binary pairs, class-mean dictionary	0.560	0.675	2.00	1.000	1.000
Deep + pair-NNLS hybrid rerank	Deep	Global residual/prior fusion, alpha 0.15	0.953	0.977	2.00	1.000	1.000
Similarity-supervised deep regressor, baseline_corrected	Deep	Top-2 margin + similarity-weighted false-compound loss	0.952	0.974	2.00	1.000	1.000
Deep coefficient regressor, baseline_corrected	Deep	Synthetic binary mixtures, top-2 support inference	0.950	0.973	2.00	1.000	1.000
CNN encoder, baseline_corrected	Deep	class mean decoder	0.879	0.937	2.00	1.000	1.000
Replicate-aware decoder, baseline_corrected	Deep	replicate atom decoder	0.852	0.923	2.00	1.000	1.000
Deep coefficient regressor, raw	Deep	Synthetic binary mixtures, top-2 support inference	0.726	0.808	2.00	1.000	1.000
Similarity-supervised deep regressor, raw	Deep	Top-2 margin + similarity-weighted false-compound loss	0.662	0.743	2.00	1.000	1.000
CNN encoder, raw	Deep	class mean decoder	0.638	0.802	2.00	1.000	1.000
Replicate-aware decoder, raw	Deep	replicate atom decoder	0.609	0.766	2.00	1.000	1.000

The table includes the main comparable classical and deep runs saved under `Unmixing_Pipeline/Results`.

“Existing” refers to the original real-mixture dataset from `mixtures_dataset.csv`.

“PT2” refers to the later real-mixture dataset defined by `Data/pt2/Mixtures.txt` and the corresponding spectra folders.

The clean classical benchmark is the replicate-aware pair-NNLS row. The diagnostic classical ceiling is the family-fallback row.

The strongest clean deep family is the baseline-corrected MLP coefficient-regressor line and its similarity-supervised follow-up.

Table 2: Per-chemical precision, recall, and F1 on the eight original supported chemicals for the main classical and deep contenders

Chemical	Clean classical			Classical ceiling			Deep baseline			Deep supervision			Hybrid		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
1-dodecanethiol	1.000	0.967	0.983	1.000	1.000	1.000	1.000	0.951	0.975	1.000	0.942	0.970	1.000	0.947	0.973
6-mercapto-1-hexanol	1.000	0.667	0.800	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
benzene	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
benzenethiol	0.667	1.000	0.800	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
etoh	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
meoh	1.000	0.959	0.979	1.000	0.963	0.981	1.000	0.922	0.959	1.000	0.934	0.966	1.000	0.942	0.970
n,n-dimethylformamide	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
pyridine	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

“Clean classical” is the frozen replicate-aware pair-NNLS benchmark.

“Classical ceiling” is the localized family-fallback variant and is included as an engineering upper bound rather than the main scientific benchmark.

“Deep baseline” is the baseline-corrected deep coefficient regressor.

“Deep supervision” is the baseline-corrected similarity-supervised follow-up.

“Hybrid” is the baseline-corrected global deep-plus-pair-NNLS reranker.